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(54) Title: AHI COMPETENCY-BASED EDUCATION PORTAL

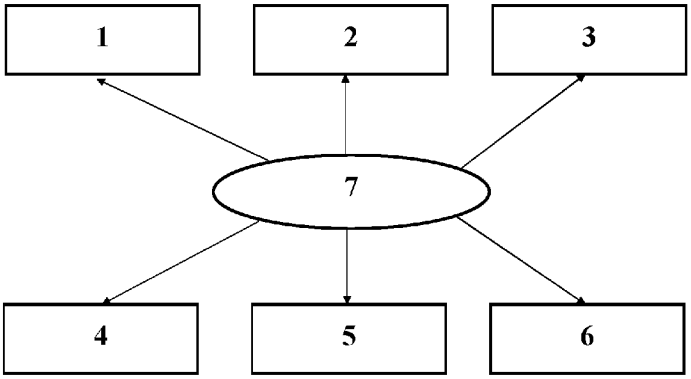

Figure-1

(57) Abstract: This invention is related to the education portal based on high competence, which aims to increase the quality of educational activities carried out especially in higher education institutions, to create a strong link between program competencies and course learning outcomes, and to establish standards in determining competencies characterized in that it comprises of; user management and authorization module (1), where authorization is performed in the system for each user through the main server, curriculum development module (2), in which the basic parameters of all academic units open in the educational institution (such as the department, department and programs in the Institute / Faculty / Schools) and the curriculum created for each program are automatically transferred to the system via the server, content definition module (3) in which the program qualifications and course learning outcomes are defined in the database of the system for academic programs, files and information packages are uploaded for the acquisitions and gains, the subject, objectives and resources of the course are defined, measurement and evaluation module (4) that allows the system to analyze

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the items (difficulty, discrimination, etc.) of the questions as a result of the analysis of the answers entered by the students into the database as a result of the exam conducted on the system and to allow the inappropriate questions to be made passive in order not to be used again by the administrators.

AHI COMPETENCY-BASED EDUCATION PORTAL

Field of the Invention:

This invention is related to the education portal based on high competence, which aims to increase the quality of educational activities carried out especially in higher education institutions, to create a strong link between program competencies and course learning outcomes, and to establish standards in determining competencies.

State of the Art:

There is no requirement to create a hierarchical connection between the National Qualifications Structure for Higher Education in Turkey (NQF-HETR) program competencies and course learning outcomes in current education systems, and there are no frameworks in place to monitor and manage this relationship. In summary, despite the lack of requirements for competency-based education in Turkish higher education and the processes by which these standards will be applied, there is no system in place to track and assess whether students have achieved the learning outcomes of the courses they have taken. Although the Bologna process suggests that competencies be considered during the preparation, implementation, and assessment stages of education, the monitoring and follow-up of this process was difficult, and it was achieved on the lecturers' initiative. The most important drawback to existing practices is that higher education credentials cannot be calculated using both pedagogical and andragogical concepts, as well as a taxonomic hierarchy. On the other hand, it is unclear if the expected competencies have been met, and no standard framework exists in this regard. There has never been a mechanism in place to determine the standard of higher education. Academicians determine the achievements of the courses they will teach and their relationship with the program adequacy and effective control of these is not included.

In today's conditions, it is very difficult to follow and control educational activities carried out in higher education institutions without a specific system. Within the scope of Bologna process applied in higher education institutions, higher education academic evaluation and quality improvement commission (YÖDEK) and National Qualifications Framework were determined. In this way, a system based on competencies has been envisaged in higher education programs, but mechanisms that will ensure the relationship and harmony between program competencies and course learning outcomes could not be established. Moreover, no system has been developed to control this relationship. However, qualifications and learning outcomes should be defined in a way that they are related to each other in higher education institutions that train specialists in certain fields and education and evaluation should be done accordingly.

The main problem in current systems is the uncertainty about achieving the competencies that students are expected to gain in programs applied in higher education. In this context, it is the absence of a Learning Management System that measures at what level the education provided in higher education institutions meets the specified competencies and the lack of a platform for the implementation of distance education. Some of the applications and deficiencies in the literature for this purpose are given below.

Application TR201903936 issued on March 15, 2019, discloses a performance enhancing quality management system. The present invention is a performance enhancing quality management system that records the information of which indicators each unit is responsible for by regularly recording the goals, objectives and indicators of each organization's strategic plans, which have strategic goals and plan to progress according to these goals. Assigning the indicators that they are responsible for to each unit, the purpose and goal definition processes are made, after the report consisting of the entered data is received, the data entries of the indicators are ensured to be closed to the users when requested for 12 months in order to prevent the error caused by the change of the data entered by the unit, personnel assignment procedures are made, the main page includes the unit-specific notifications and menu titles according to the authorization status, the data entered are analyzed in tables according to the period

selected in the source, using the formulas defined for each indicator and advanced reporting techniques, the last three data entered are monitored with automatic graphics a system is discussed.

Application TR201911609 issued on July 31, 2019, discloses a system that provides measurement and evaluation by reaching education and program outcomes The present invention is that it can be integrated into student information automation systems in educational institutions or can be used externally, it is ensured that the learning outcomes that express the qualifications and competencies that students need to gain in a certain course at the end of the education given in educational institutions can be reached online, the student will have at the end of the program. Knowledge, skills and competencies are defined, program outcomes are accessed online, the exam questions prepared by the instructor are matched with the data in the learning output module to which these questions are related, the questions are matched with the data in the learning output module, and the questions belonging to the students over the exam result from the optical reader in multiple-choice exams Include all data at the end of the education program, which is transferred to the database by creating and transferring data-based success data, statistics graphics are created by comparing the data and these graphics are delivered to the instructor in the relevant department. It is discussed a system where a general report is created and that offers solutions to the users for student or course-based improvements.

Application US10515562B2 issued on May 4, 2017, discloses a systems and methods for instrumentation of education processes. Systems and methods for education instrumentation can include one or more servers configured generate a plurality of models for modeling various aspects of an education process using training data related to academic performance of students. The one or more servers can collect data from client devices associated with various education institutions or stakeholders throughout an life cycle of the education process. The one or more servers can use the generated models and the collected data to assess the addressing of education standards and predict or estimate performance metrics associated with the education process. The one

or more servers can provide computed metrics or assessments of how well education standards are addressed to one or more client devices for display.

The above-mentioned applications for quality improvement control have been addressed. During the evaluation process, though, there is no evaluation by academic staff of these applications. Expert access and inspection are not possible with such devices.

Consequently, there is a necessity for a novel technology that is capable of eliminating above mentioned disadvantages, increase the quality of educational activities, create a strong link between program competencies and course learning outcomes, and set standards in determining competencies.

Description of the Invention:

The present invention is an ahi competency-based education portal increase the quality of educational activities, create a strong link between program competencies and course learning outcomes, and set standards in determining competencies.

The main purpose of the invention is to design, develop and put into use a Learning Management System software that can be used in the human resource training and development process. With the invention, it is aimed to increase the quality of educational activities carried out in higher education institutions, to create a strong link between program competencies and course learning outcomes, and to set standards in determining competencies. The following are the basic realizations envisaged by the invention in conjunction with this key purpose:

- To guarantee the learning rights of learners,
- To build educational programs on a sustainable and inclusive understanding that will bring the values of ahi-order based on humanitarian virtues,
- Achieving success for the benefit of the student in determining the program competencies, making the courses standardized in the name of the curriculum and gaining the course competencies,

- Establishing mechanisms in accordance with transparency and audit principles in planning and evaluating training processes,
- Defining the subjects, learning outcomes and achievements of each course using the inventive system, establishing the relationship of the subjects with the learning outcomes,
- To make an assessment in accordance with the gains in the course evaluation phase and to determine the acquisition of the course's competencies
- Developing a Distance Education Management System that can be used in emergency situations such as natural disasters and epidemics, to ensure that education is not interrupted,
- Sharing files for the subjects and achievements of the lessons in the distance education process,
- To ensure effective communication between students and instructors in the distance education process,
- To ensure the integration of the quality assurance system based on competence in education applied at the university with external quality assurance systems through accreditation,
- To shape the design of training programs with a competence-based understanding based on national and international standards and values,
- Developing a unique Ahi Education Quality Assurance System Model to be followed in the design, planning and execution of educational processes,
- To build a new competency-based accreditation model that is exemplary at national and international level.

It will be possible to enhance the quality of higher education institutions by introducing an auditable and sustainable structure that can plan, execute, and assess education and training activities in the higher education system within the context of the developed standards, as a result of the above-mentioned objectives.

Education portal based on the subject of invention has been developed to ensure that education processes in higher education institutions are carried out in accordance with certain standards and to create a quality assurance system in education. The invention

ensures that the planning, implementation and evaluation processes related to education are carried out in accordance with certain standards.

With the inventive system, the acquisition status of the competencies foreseen in education systems is measured as learning outcomes. Follow-up is provided in accordance with the standards in competency-based training. Thanks to the developed system, competencies are determined in a hierarchical order and all the anticipated gains serve the program competencies.

Based on the competency-based education approach, the invention realizes all processes related to education through an internet-based learning management system.

The basic principles of the inventive system are as follows:

- It guarantees the right to learn.
- It focuses on learning and the multidimensional development of the student.
- It is based on student-centered education.
- It is aimed to gain predetermined competencies.
- It is based on the harmony between the program competencies and the learning outcomes of the courses.
- It is based on the harmony between the learning outcomes of the courses and teaching practices.
- It is based on providing rich teaching material and interactive teaching environment.
- It is based on the use of measurement and evaluation model for the qualifications and learning outcomes of the courses.
- It is based on making a joint examination for the same course.
- Aims to evaluate programs and their continuous improvement.
- It directs the lecturers conducting the same course to make decisions and cooperate together in order to gain competencies.
- Provides standardization in the acquisition and assessment of the courses.
- It increases the academic interaction of lecturers working in the same unit in higher education institutions.

- It makes it necessary to create an up-to-date curriculum for the courses in higher education programs.

The invention has a distinctive structure and in many ways has advantages. The following are some of the invention's benefits:

- It enables the establishment of a spiral (holistic) structure between the programs, curriculum and courses in higher education.
- In higher education, it allows all courses to be standardized in its institutions.
- Thanks to the support offices and commissions established in academic units within the scope of the invention, it enables the development of program competencies and course learning outcomes with a holistic and interdisciplinary perspective.
- It allows assessment and evaluation appropriate to the outcomes and eliminates subjective evaluations based on the initiative of the instructor.
- It can be determined to what extent students have reached the planned achievements.
- Administrators, lecturers and students can follow all education-related processes through the system.
- Accessible via common information technologies (computer, tablet and mobile phone),
- Thanks to its ability to be used independently of time and place, it enables uninterrupted education.
- It can be integrated with different video conference systems and can be used for distance education.
- It enables the creation of course syllabuses in the face-to-face education process.
- Bologna Information Package can be created automatically as a result of the information defined on it.
- It allows data sharing with Student Information Systems used in higher education institutions, automatically pulls basic data from this system and updates it continuously.
- Unlike other learning management systems, it has a competency-based structure. Thus, in higher education programs, it creates competency-based hierarchical

charts and provides the opportunity to monitor the acquisition of targeted competencies.

- Thanks to the invention, students can see the program competencies, course learning outcomes and achievements in advance, and they are informed about the preparation for the lesson.
- An effective communication can be established between the instructors and students through the system.
- Competence-based measurement and evaluation is carried out through the system.
- The system can be easily updated according to needs.
- Keeping the contents to be created for the lessons for a long time provides the advantage of creating a pool of learning material.

All the advantages of the product subject to the invention will be understood more clearly thanks to the scheme given below and the detailed explanation written by making references to this scheme, and therefore the evaluation should be made by taking this scheme and detailed description into consideration.

Brief Description of the Drawings:

The invention will now be described with reference to the accompanying figures, so that the features of the invention will be more clearly understood and appreciated, but not by limiting the invention to these particular embodiments. On the contrary, it is intended to cover all alternatives, modifications and equivalents that may be included within the scope of the invention as defined by the appended claims. It should be understood that the details are shown only for the purpose of illustrating preferred embodiments of the present invention and are provided the most useful and easily understood descriptions of both the method and the rules and conceptual features of the invention. In these drawings;

Figure 1 That is the schematic view of the components comprising the innovation framework.

The figures that will help to understand the present invention are numbered as indicated in the attached picture and are given below with their names.

Disclosure of References:

1. User Management and Authorization Module
2. Curriculum Development Module
3. Content Definition Module
4. Measurement and Evaluation Module
5. Online Course Module
6. Tracking and Attendance Module
7. Course Registration and Monitoring Module

Disclosure of the Invention:

The present invention is comprised of user management and authorization module (1), where authorization is performed in the system for each user through the main server, curriculum development module (2), in which the basic parameters of all academic units open in the educational institution (such as the department, department and programs in the Institute / Faculty / Schools) and the curriculum created for each program are automatically transferred to the system via the server, content definition module (3) in which the program qualifications and course learning outcomes are defined in the database of the system for academic programs, files and information packages are uploaded for the acquisitions and gains, the subject, objectives and resources of the course are defined, measurement and evaluation module (4) that allows the system to analyze the items (difficulty, discrimination, etc.) of the questions as a result of the analysis of the answers entered by the students into the database as a result of the exam conducted on the system and to allow the inappropriate questions to be made passive in order not to be used again by the administrators.

The user management and authorization module (1) of the invention is characterized by grouping users over the main server and granting them collective authorization.

The user management and authorization module (1) of the invention is characterized by authorization transition for users with different authorizations on the main server.

Measurement and evaluation module (4) of the invention is characterized by performing success analysis of the class or group determined as a result of performing data analysis on the processor during the active period.

Measurement and evaluation module (4) of the invention is characterized by suggesting exam questions to the department responsible for the exams assigned by the user management and authorization module (1) as a result of the analysis

Measurement and evaluation module (4) of the invention is characterized by providing instructors to create a question pool by entering questions appropriate to their acquisitions into groups categorized for each course on the system.

Measurement and evaluation module (4) of the invention is characterized by reporting in the database all the evaluations that students have been subjected to throughout their education life, showing the degree of achievement of the qualifications determined at the graduation stage.

Measurement and evaluation module (4) of the invention is characterized by users who are authorized to create a measurement and evaluation module (4) for the courses, and consent to the authorization to conduct exams.

The user management and authorization module (1) of the invention is characterized by transmitting the entered data to the relevant person or persons over the server in order to control all definitions entered into the system's database and submit it to expert evaluation.

Detailed Disclosure of the Invention:

This invention is a Learning Management System used in educational institutions. This system Learning Management System consists of various modules. The modules and functions that make up the invention are:

1- User Management and Authorization Module (1): The system subject to the invention works on the basis of authorization. Authorization can be made for each user by the system administrator. Authorization can be canceled or added when necessary. Users can be grouped and collectively authorized. Users with different authorizations can easily switch to different authorizations within the system. Table view of the module is as follows.

Table-1

Authorities Belonging to You	
Head of Department/Program	✓
Unit AYDEP Coordinator	✓
Head Of Department	✓
Dean / Director	✓
Assistant Dean/Director	✓
Secretary General	✓
Head of System Support Office	✓
System Administrator	✓
YOK Assessment	✓
Head of Evaluation Assessment Support Office	✓
Student	✓
Instructor	✓
Head of Instructional Technology Support Office	✓

2- Curriculum Development Module (2) and Content Definition Module (3): It is ensured that the basic parameters of all academic units open in a higher education institution (department, department and programs in the Institute / Faculty / High Schools) and the curriculum created for each program are automatically transferred to the system via the server. Students who take each course are automatically defined to the system.

The curriculum development module (2) and content definition module (3) of the inventive system allow to define program competencies and course learning outcomes for academic programs, to download files and information packages for their outcomes and gains, and to define the subject, objectives and resources of the course. The sample table view of the definition module (3) in which this academic unit data is entered is as follows.

Table -2

Page 10 Search :

#	Departments	Head Of Department	Number of Departments	Number of Lessons	Number of Branches	Number of Programs
	Computer Technology (2.1)	Research Assistant [illegible]	2	72	26	5
	Plant and Annual Production (2.2)	Research Assistant [illegible]	3	90	48	5
	Office Services and Secretariat (1)	Research Assistant [illegible]	2	118	58	3
	Veterinary Medicine (1)	Ph.D. Faculty Member [illegible]	3	52	24	2
	Child Care and Youth Services (1)	Research Assistant [illegible]	3	54	27	2
	Accounting and Tax (1)	Ph.D. Faculty Member [illegible]	3	18	9	2

Show 1 to 6 of 6 entries

Learning outcomes and acquisitions entered into the inventive system are submitted to the approval of the program development and evaluation support office and the relevant field experts defined by the user management and authorization module (1) in the system, and necessary corrections are made.

A sample Curriculum Development chart based on competence is provided below:

Table -3

PROGRAM	PROGRAM COMPETENCES			CLASSES	SUB-JECTS	LEAR-NING OUTCO ME	ACHIEVE MENT	QUESTIONS
	Competence Area	Competence Definition	Competence Indicator					
Program A	Professional Knowledge	At least 3	At least 2	B Class	14 weeks	At least 2 for each week	At least 2 for each Learning Outcome	At least 2
	Professional Skill	At least 3	At least 2			At least 2 for each week	At least 2 for each Learning Outcome	At least 2
	Attitude and Value	At least 3	At least 2			At least 2 for each week	At least 2 for each Learning Outcome	At least 2

In the curriculum development module (2), competency areas have been created specifically. Competence areas are Professional Knowledge, Professional Skill, Attitude and Values. While defining the program competencies, the competency area is definitely selected and the necessary definitions are made by the content definition module (3).

The following operations are carried out in the curriculum development process carried out by the curriculum development module (2) and the content definition module (3):

a) Introducing the instructors who teach the course, the experts who will supervise the course and the students to the system,

b) The course and the instructor are matched, and the instructor determines the subject of the course and the learning outcomes of the course in accordance with the program competencies and enters the course in accordance with the taxonomic characteristics of the course outcomes

c) Uploading the information package for the gains,

d) Teaching staff to create the competency matrix that relates learning outcomes and program competencies,

e) Submission of the table of indications prepared by the instructors to expert opinion,

f) Evaluating the course learning outcomes and gains entered into the system by field experts and lecturers and giving feedback about the deficiencies,

g) Making necessary corrections in the system by the instructor in line with expert opinions,

h) Approval of the indicator table by the experts.

The sample table view of the tabs of the curriculum development module (2) on the portal is as follows:

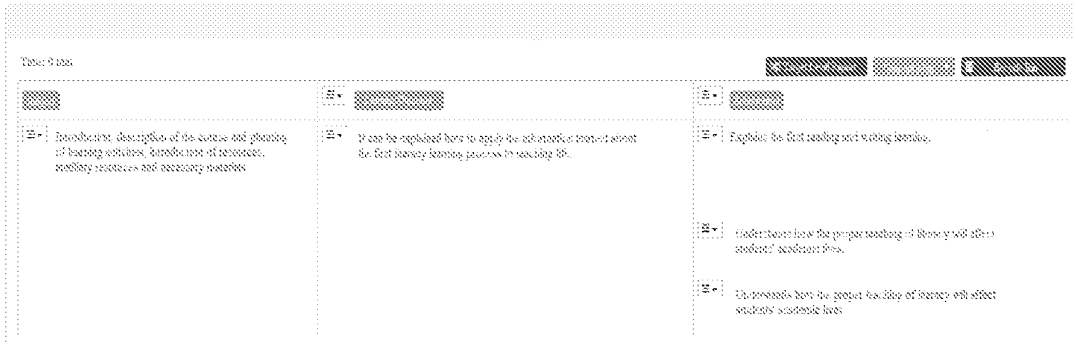
Table -4

First Reading and Writing Learning (20132112013) — Subject

§	Course Subject
	Introduction of the course, planning of teaching activities, introduction of resources, auxiliary resources and necessary materials.
	Foundation and evaluation of the parts of the 2019 Turkish Program related to the First Literacy Teaching.
	Aims and principles of first reading and writing teaching, Features of Turkish, the effect of these features on teaching reading and writing.
	The basic characteristics of first year students, Reasons of inefficiency before in first reading and writing teaching.
	Tools and materials used in primary reading and writing teaching (look features and effects, selection, creation and use of these tools).
	Methods (definition, characteristics, classification, applications, advantages and limitations of methods applied in our country and in the world in primary reading and writing teaching.
	Sound-based sentence method (definition, principles, features, stages and applications). Principles of preparing presentations for application studies.
	Sample application studies of Sound Based Sentence Method. Exercising for activity examples and preparing activities suitable for the first reading and writing teaching stages.
	Presentations and discussions on teaching reading and writing practices in line with the stages of the sound-based sentence method (1st group letters: evaluation of teaching a-ha-l-on letters).
	Presentations and discussions on teaching reading and writing practices in line with the stages of sound-based sentence method (1st group letters: evaluation of teaching [illegible] letters).
	Presentations and discussions about teaching practices of reading and writing in line with the stages of the sound-based sentence method (1st group letters: evaluation of teaching the letters [illegible]).
	Presentations and discussions about teaching practices of reading and writing in line with the stages of the sound-based sentence method (1st group letters: evaluation of teaching the letters [illegible]).
	Presentations and discussions about teaching practices of reading and writing in line with the stages of the sound-based sentence method (1st group letters: evaluation of teaching the letters [illegible]).
	Second evaluation: presentations and discussions about reading and writing teaching practices in line with the stages of the sound-based sentence method.

The sample table view of the tabs of content definition modules (3) on the portal is as follows:

Table -5



3- Measurement and Evaluation Module (4): By using the inventive system, assessment and evaluation can be made in 6 different types (multiple-choice, online written, online oral, performance assignment, objective structured performance exam).

The sample table view of the tabs of the measurement and evaluation module (4) is as follows:

Table -6

The measurement and evaluation module (4) can determine the date, time and duration for online exams in the system, allowing the opportunity to change the date and time and increase the duration. Reading and scoring online test exams can be done automatically. For homework-based evaluations, homework can be uploaded, commented on uploaded assignments, and re-uploaded files. Evaluation scales (rubrics)

are prepared for homework and written evaluations and evaluation can be made accordingly. The sample table display of the exam analysis result is as follows:

Table -7

EXAM QUESTION ANALYSIS

Exam Name: Basic Statistics (Midterm)

Exam Date : 04/02/2020

Exam Question Analysis

Number of Students Taking the Exam : 108

Number of Lower and Upper Group Students: 29

Number of Questions : 14

#	Root of The Question	Question Type	Options					Success Rate	Question Owner	Difficulty	Distinctiveness
1	.....	Multiple choice	*	Upper G. C.S	Lower G. C.S	Total Lower/Upper G. C.S	All Students C.S.	96	Dr. Öğr. Üyesi AHMET SALİH ŞAHİN	So hard (0.00)	Must not be used (0.00)
			.....	0	1	2	6				
			.....	29	16	45	20				
			.....	5	3	3	4				
			.....	0	4	1	2				
			.....	0	1	1	2				
2	.....	Multiple choice	*	Upper G. C.S	Lower G. C.S	Total Lower/Upper G. C.S	All Students C.S	96	Dr. Öğr. Üyesi AHMET SALİH ŞAHİN	So hard (0.00)	Must not be used (0.00)
			.....	0	6	6	7				
			.....	26	12	42	87				

The measurement and evaluation module (4) of the invention also enables the evaluation of homework and the entry of the grades into the system. Optionally, students' evaluation results and justifications can be viewed here.

The system, which is the subject of the invention, performs item analysis (difficulty, discrimination, etc.) of the questions as a result of the analysis of the questions used in the exam through the answers entered into the database by the students, and allows inappropriate questions to be deactivated in order not to be reused by the administrators.

Achievement analyzes of the class can be made by the measurement and evaluation module (4) of the system of the invention as a result of processor data analysis. The questions prepared are submitted for approval by the Assessment and Evaluation Support Office and the relevant field specialist, and accepted after necessary corrections.

Assessment and evaluation module (4) creates a question pool consisting of questions suitable for the outcomes by the instructors. The exams to be made can be created by using the questions in this pool.

The instructor prepares a rubric for assignments, defines requirements, start and end dates for the assignment and shares it with students. The steps followed in the process of creating and conducting exams on the system are as follows:

a) Entering questions for assessment and evaluation of course outcomes by instructors,
b) Sending the created exam questions to experts determined by the Assessment and Evaluation Commission,
c) Checking the conformity of the questions prepared by experts with the gains and giving feedback to the instructor,
d) Making necessary corrections by the instructor and sending them for approval again,
e) Approval of exam questions by experts,
f) Defining the exam consisting of approved questions to the system before the exam,
g) Appointment of supervisors for exams,
h) Conducting the exam on the system at the specified date and time,
i) Removing inappropriate questions from the system based on the item analysis calculated by the measurement and evaluation module (4) for the questions after the exam.

The competency-based measurement and evaluation module (4) in the system of the invention provides evidence showing the degree to which the students have achieved the qualifications determined at the graduation stage by storing all the evaluations they have been subjected to throughout their education life in the database.

Throughout the academic year, exams can be carried out in the form of electronic exams on the system, and the instructor can print the exam he has created in Word format and apply it in the classroom. In electronic exams, the instructor can determine the duration

of the exam and extend the time when necessary. At the end of the exam, students can see their grades. If requested, students are provided to see the mistakes and correctness in the exam they entered. With the rubric prepared for homework, students can see which answers and how many points are given by the instructor, and they can comment on the assignments.

4- **Online Course Module (5):** Live lectures and meetings can be held thanks to the feature of integrating a web-based video conference system into the system, which is the subject of the invention. Online live lectures and meetings can be created through the system. Thesis defenses, seminar presentations and academic meetings and events are defined and participation of authorized persons can be ensured. Operations such as removing participants from the system, silencing, and authorizing presentation can be performed upon request.

Video or audio participation can be provided in online lessons or events. The session authority can send bulk or private messages to the participants. A file can be created for the decisions or opinions to be taken during the session, and opinions and suggestions can be written without any intervention. The session official can make changes on the shared notes when necessary. Surveys can be organized during the online course or meeting. Authorized unit officers can instantly monitor the realization of planned lessons and can attend online lessons when necessary. They can create reports on the attendance rates of the courses.

5- **Tracking and Attendance Module (6):** Reports showing the attendance status in courses or other activities created over the inventive system can be obtained. Retrospective information about the courses and the participating students can be obtained. Reports that can be received in the follow-up and attendance module (6):

a) Faculty members can get attendance reports for their courses.

b) Students can view the reports of their attendance to the courses.

c) Managers can view the course attendance reports.

6- Course Registration and Monitoring Module (7): It allows recording of live lectures online. The recorded lessons can then be opened to the students to watch within the specified time interval.

The invention allows the development of all modules and making the necessary data updates. Thus, it facilitates data-based updates for the courses and programs they are responsible for.

The system subject to invention includes some new principles and innovative and functional methods in accordance with these principles in the design, implementation and evaluation of educational activities in higher education processes. The new methods introduced by the system and explanations regarding the functions of these methods are given below.

System Login Method: Login to the system requires the same information as entering the Student Information System. Faculty members can log in with their registration numbers and passwords, and students with their student numbers and passwords. Login to the system is made according to the authorization method.

Authorization Method: When the system is first entered, the defined tasks screen appears, and the system login is completed by selecting the task. There are many authorities within the system according to their functions. Users can log in to the system by choosing their authorization according to the operation they want to do. The authorizations are in line with the authorities defined in the legal regulations of higher education institutions and existing in student information systems. In this context, there are authorities in accordance with the administrative hierarchy of academic units (For example: lecturer, student, head of department, head of department, dean, director, academic advisor, system administrator, office or commission chairman, etc.).

Definition Method: The system allows to make definitions regarding the design, implementation and evaluation of educational processes. In the identification process, basic information (academic year, semester, unit, program, curriculum, courses and

students and course managers) is automatically retrieved from the Student Information System. New definitions can be made manually on the system when necessary. The information in the Student Information System is constantly updated within defined time intervals. In addition, updates can be made by the authorities at any time. The definitions to be made in the system are made according to the principles of authorization (For example: while program competencies can only be made by the Head of Department / Department, course learning outcomes can only be made by the instructor in charge). Instructors write the learning outcomes / achievements for each course they are responsible for and associate them with the relevant program outcomes. Instructors write the 14-week topic list of the course and associate it with the learning outcomes. Instructors can also enter the acquisitions associated with course learning outcomes in accordance with the target area and target level in the Bloom taxonomy. As a result of these definitions, a table of indications is formed for each course. Again, by the instructors, the student workload, resources, measurement and evaluation methods and tools, tasks and activities etc. definitions are made. As a result of these definitions, Bologna Definitions, which are mandatory in universities, are formed and can be taken as a report. The Identification Method in the system not only defines the competency expectation targeted within the scope of Bologna, but also ensures its compatibility with the application and evaluation processes. Instructors can define the purpose of the course, source, file, forum, feedback and contact information for the courses they are responsible for. An important function of the system in terms of process tracking and monitoring is that it can create an assessment and evaluation module (4) for the course. Because, an exam cannot be held before the course assessment and evaluation module (4) is created. With the course assessment and evaluation module (4), the instructor indicates the percentage weights and numbers of the exams he will conduct in terms of assessment. The questions to be determined for the exam are determined in accordance with the outcomes of the courses. These definitions for evaluation purposes are intended to establish the assessment and evaluation system at the beginning of the semester and to inform students.

Process Tracking and Monitoring Method: All kinds of data / information / task defined in the system can be seen by the users in line with the authorization principles.

With this method, students can see the subjects, learning outcomes, achievements and the resources and files uploaded for the courses they take. In addition, they have the authority to see and follow the assignments given by the teaching staff, the announcements made and the forms opened. Instructors can perform actions such as taking attendance, seeing and evaluating the assigned assignments, giving feedback, and sharing announcements so that they can monitor the attendance to the lesson. It is examined by experts in terms of content validity.

Online Lecture and Meeting Method: The system allows various activities to be held within the scope of distance education activities. In this context, online lessons and meetings can be organized. In the use of this method, it is possible for people authorized to create lessons or meetings whenever they want. Online lectures and meetings held with video conferencing software integrated into the system are recorded and allowed to be followed later. 100 online lessons / meetings can be organized instantly through the system, and 200 people can participate in each course / meeting without any problems. With this method, video or audio online activities can be done. During the online event, collective or individual messaging, file sharing, presentation authorization, screen sharing can be done. Participants can be restricted or authorized to use functions and / or canceled by the administrator in online courses / meetings.

Supervision, Assessment and Support Method: The system has a structure that removes the critical definitions that direct the learning-teaching process from the initiative of the instructor. All definitions made in accordance with the basic principles guiding the establishment of the system are open to audit and are presented for expert evaluation. The inspection and evaluation processes carried out in the system subject to the invention are carried out by the support units. Within the scope of the invention, support commissions for curriculum development and measurement and evaluation have been established in all academic units within the scope of the definitions regarding instructional processes. The commissions evaluate the definitions in their respective areas of authority and provide feedback to the lecturers through the system. Support management ensures transparency and control of educational processes; It allows development with an interdisciplinary perspective. With the demand section on the

system, academic staff can receive support from system administrators regarding the problems they experience. In addition, there is a user manual about the general features of the system.

The coordinator, support offices and commissions listed below are required for the implementation of the competency-based education quality assurance system model and the implementation of the portal and learning management system designed within the scope of this model. The functions of the mentioned coordinator, support offices and commissions are as follows:

Quality Assurance System Coordinator in Education: A Quality Assurance System in Education (QASCE) is established within the Rectorate and coordinates all other support offices and commissions. The duties of the Quality Assurance System Coordinator in Education within the scope of the invention are as follows:

a) To carry out planning, implementation, monitoring and improvement studies to increase the quality of educational processes and practices.

b) To prepare and execute action plans for the development of a quality assurance system based on competence in education.

c) To work on the execution of educational activities to be carried out through formal or distance education in accordance with the principles of the Quality Assurance System in Education and to ensure that other units are assigned or authorized in matters related to their field of duty

d) Coordinating the accreditation processes of the Departments / Programs.

e) To develop suggestions for conducting education-training activities based on competence

f) To develop Learning Management Systems for competency-based education in cooperation with UZEM and to make these systems widespread.

g) Coordinating the commissions and support offices established for the healthy execution of learning management systems and determining their working principles and procedures.

h) To convey and follow up requests, suggestions and complaints from academic units and other stakeholders regarding the development of learning management systems to the relevant coordinator, office, board and persons.

i) To ensure the functioning and coordination of the unit coordinators, commissions and support offices to be established within the scope of the competency-based quality assurance system in education in line with the determined principles

j) Notifying the decisions taken within the scope of the quality assurance system to the relevant coordinator, support office and individuals and following up the decisions.

k) Conducting research on the quality of education and educational processes and presenting the research results to the Rectorate and sharing them with the relevant units.

l) To communicate with national and international independent accreditation bodies authorized for the accreditation of academic programs and to coordinate the necessary information and training activities during the accreditation process.

m) To determine the processes, activity plans and performance parameters for the Quality Assurance System Based on Competence in Education.

n) To carry out joint activities and develop projects with universities and other institutions / organizations in Turkey and abroad in order to achieve excellence in education and training activities.

Curriculum Support Office: Under the coordination of the quality assurance system in education, a curriculum support office (CSO) is established and its duties within the scope of the invention are as follows:

a) The adequacy of the academic units of the programs and curricula of National Qualifications Framework for Higher Education in Turkey to report on compliance.

b) To provide training to unit managers and lecturers on the quality assurance system based on competence in education, competency-based education and curriculum development.

c) Preparing learning outcomes and specification tables, preparing a guide on the issues related to the formation of the program competencies matrix.

d) To do the necessary monitoring and guidance studies for the learning management systems used by the units to be competency-based.

e) To guide the curriculum development and evaluation commissions to be established in the units.

Measurement and Evaluation Support Office: A measurement and evaluation support office (MESO) has been established under the coordination of the quality assurance system in education and its duties within the scope of the system subject to the invention are as follows:

a) To ensure that educational measurement and evaluation procedures and processes are carried out in accordance with the principles of the quality assurance system based on competence in education and the understanding of competency-based education.

b) To train instructors in the development, application and scoring of measurement and evaluation tools and techniques, writing questions, and preparing rubrics (graded scoring).

c) To monitor the appropriateness of measurement and evaluation tools and methods to the content, scope, gains and learning outcomes of the course and to contribute to its development.

d) To give feedback to the instructors regarding the validity, reliability and usefulness of the measurement tools and techniques, upon request.

e) To share the decisions and suggestions made by the quality assurance system coordinator regarding the measurement and evaluation processes with the instructors.

f) Planning and conducting trainings for unit managers and lecturers on competency-based measurement and evaluation in accordance with the principles of the quality assurance system based on competence in education.

g) The functioning of the competency-based measurement and evaluation system, the principles of practice, effective and up-to-date approaches regarding the

measurement and evaluation processes and the preparation of questions, etc. To prepare a guide on the subjects and to inform the instructors when necessary.

h) To make suggestions regarding the measurement and evaluation interface and functions of Learning Management Systems.

System Support Office: A system support office (SSO) is established under the quality assurance system coordination in education and its duties within the scope of the invention are as follows:

a) To take the necessary precautions and ensure the necessary coordination with the relevant person / unit / parties in order for the learning management systems to work technically smoothly.

b) To convey their suggestions for the healthy execution of learning management systems to the quality assurance system coordinator.

c) Evaluating requests, suggestions and complaints from stakeholders regarding learning management systems, and conveying and following up the software developer or company officials.

d) To intervene in technical problems that may arise in the use of learning management systems.

e) To prepare written and / or visual guides for students and lecturers on the use of learning management systems interfaces.

f) To make necessary interventions to increase the usefulness of the interfaces of learning management systems and / or to follow up by conveying the software to the service provider / contractor authorities.

g) To inform the quality assurance system coordinator about the status of the requests submitted to the contractor company authorities for the development of learning management systems.

h) To inform the relevant stakeholders (students, lecturers, support offices and coordinators, etc.) about the changes and improvements made in the interface of learning management systems.

i) To train stakeholders (students, instructors, examiners, etc.) on the use of learning management systems.

Accreditation Support Office: An accreditation support office (ASO) is established under the coordination of the quality assurance system in education and its duties within the scope of the invention are as follows:

a) Coordinating the accreditation activities to be carried out for the programs.
b) To provide information and suggestions to the Unit / Department Boards about accreditation institutions and processes.
c) To prepare activity plans and reports on accreditation studies and to submit them to the coordinator of the quality assurance system in education.
d) Preparing evaluation reports on accredited Departments / Programs and submitting them to the rector's office and the coordinator of the quality assurance system in education.
e) To report to the coordinator of the quality assurance system in education on the activities carried out in cooperation with the Unit / Department Accreditation commissions.
f) Supporting national and international accreditation applications and preparing a self-evaluation report.

Academic Development Support Office: Academic Development Support Office (ADSO) is established under the coordination of the quality assurance system in education and its duties within the scope of the invention are as follows:

a) To conduct research on the development needs of academic staff and related administrative staff within the scope of the education quality assurance system.
b) To prepare and carry out an academic development program for academic staff.

Units Quality Assurance System in Education: The quality assurance system coordinator in education is established in the academic units and the unit quality assurance system in education (UQASE) is established and the tasks within the scope of the invention are as follows:

a) To carry out the educational activities in the unit in line with the principles of the quality assurance system based on competence in education and to ensure the coordination of accreditation processes.

b) To prepare an annual evaluation report on the practices for conducting education-training activities in the unit based on competence.

c) To ensure that the program qualifications, course learning outcomes and achievements of the department / department are entered into the learning management system.

d) To determine the measurable goals of the units related to education and accreditation processes, to determine the performance indicators related to these goals and to monitor them regularly.

e) To ensure the coordination of the commissions / boards to be established in the Departments for the quality assurance system and accreditation based on competence in education.

f) To prepare reports requested by the top manager of the unit on the quality assurance system and accreditation based on competence in education.

g) To convey the demands of the academic units regarding the quality assurance system and accreditation based on competence in education to the unit managers.

h) To provide suggestions for the development of learning management systems used in the unit.

i) To convey and follow up requests, suggestions and complaints received from academic units and other stakeholders for the development of the learning management system used in the unit to the relevant commissions, offices, boards and individuals.

j) Coordinating the self-assessment studies in the unit, preparing the annual unit self-evaluation report and submitting it to the coordinator of the quality assurance system and accreditation based on competence in university education at the end of March each year.

k) To establish and manage commissions when necessary in matters related to its field of duty.

l) To study and report on the evaluation of the performances of the academic staff regarding educational activities.

Curriculum Development and Evaluation Commission: In order to ensure that the educational activities carried out in the academic units are carried out in accordance with the principles of the quality assurance system, the curriculum development and evaluation commission (CDEC) established in each program with students is established and the tasks within the scope of the invention are:

a) To ensure that the programs are developed in accordance with the competency-based quality assurance system and competency-based education principles.

b) To evaluate the program qualifications entered into the learning management system in line with the principles of the quality assurance system and to inform the relevant parties about those that are not suitable

c) To assign relevant experts to evaluate the conformity of the course learning outcomes entered into the learning management system by the instructors with the program competencies and the competency-based education principles and to provide feedback to the relevant lecturer about the determined inconveniences.

d) Preparing guidelines on current approaches in program evaluation and development, learning outcomes, indicator tables, creating program competencies matrix, and organizing informative meetings on these issues.

e) To prepare reports on curriculum development and to organize meetings for lecturers.

f) Evaluating the programs by taking the opinions of students and lecturers every year, and presenting their suggestions on basic and elective courses to be added, removed, reduced or increased credits to the relevant program top managers.

g) To establish mechanisms through which stakeholders (academic staff, students, sector representatives, etc.) can comment on the current curriculum.

h) To inform the system support office of the development requests of the learning management system regarding the curriculum development and evaluation module.

Measurement and Evaluation Commission: A measurement and evaluation commission (MEC) is established in academic units in order to manage the measurement and evaluation processes in accordance with the program competencies

and course learning outcomes, and its duties within the scope of the invention are as follows:

a) To ensure that the measurement and evaluation procedures and processes in the program are carried out in line with the quality assurance system and accreditation principles based on competence in education.

b) To evaluate the conformity of the measurement tools prepared by the instructors with the quality assurance system and accreditation principles based on competence in education.

c) To support lecturers in the development, implementation and scoring of measurement and evaluation tools and techniques.

d) Writing questions for lecturers, preparing rubrics (rubrics), etc. to support measurement and evaluation activities.

e) To examine and report the validity, reliability and usefulness features of assessment and evaluation tools and techniques used in education.

f) To give feedback to the instructors regarding the validity, reliability and usefulness of the measurement tools and techniques, upon request.

g) The functioning of the measurement and evaluation system, application principles, assessment and evaluation processes, question preparation, etc. to prepare a guide and provide training on the subjects.

h) To prepare a guide about effective and up-to-date approaches in measurement and evaluation practices and to inform the instructors when necessary.

i) Evaluating and concluding the objections of the students to the exam questions in terms of the conformity with the learning outcomes and the conformity of the questions in terms of structural and formal aspects.

j) To inform the System Support Office of the development requests for the measurement and evaluation module of the Learning Management System.

Accreditation Commission: An accreditation commission (AC) is established in order to carry out the accreditation studies of the programs in each of the programs with students in the academic units and their duties within the scope of the invention are as follows:

a) To prepare activity plans and reports on the accreditation studies carried out in the relevant program and to submit them to the coordinator of the unit's quality assurance system based on competence in education.

b) To carry out the processes and procedures regarding the accreditation of programs and to prepare self-evaluation reports.

c) To prepare reports on accreditation issues at least once a year and to organize a meeting for academic staff.

d) To inform the relevant program boards about accreditation institutions and processes and to offer suggestions.

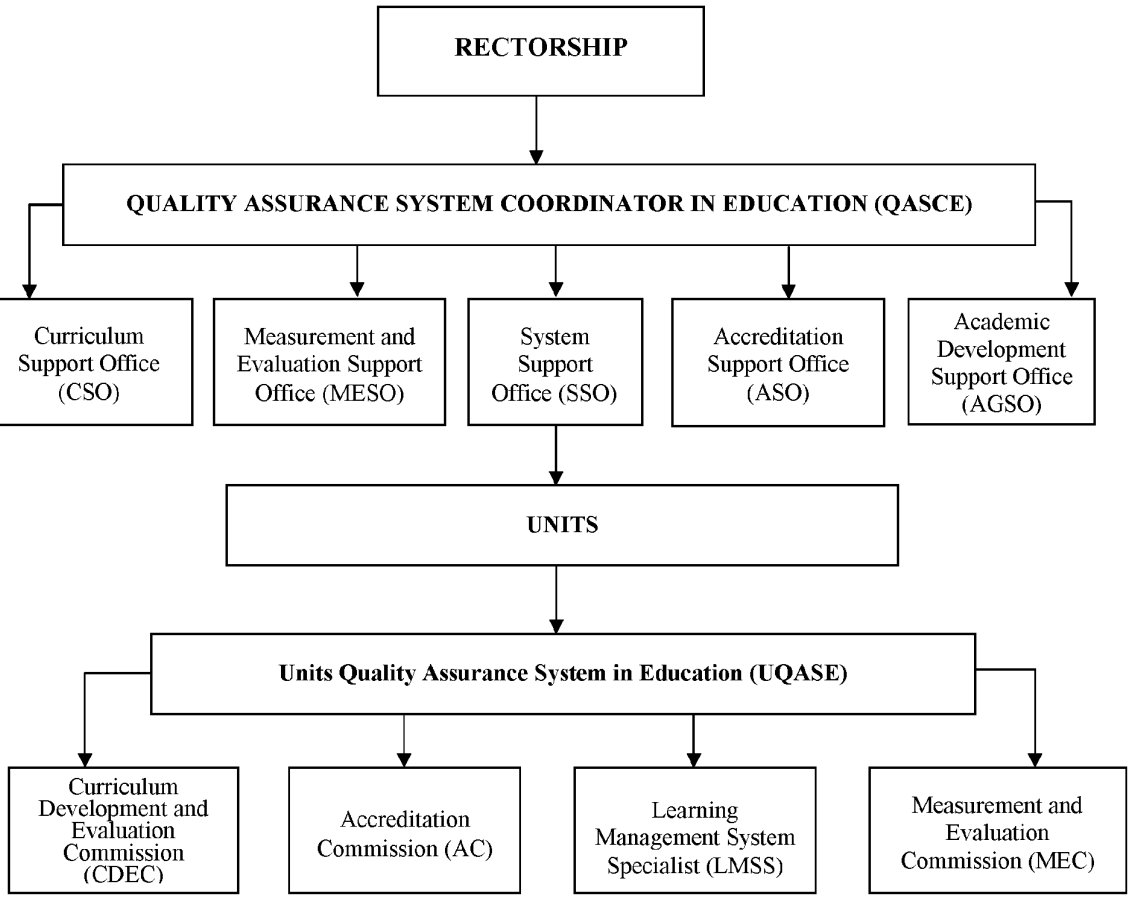
e) To inform the system support office of the development requests for the accreditation module of the learning management system.

Learning Management System Specialist: Learning management system specialist (LMSS) is appointed in all academic units and their duties within the scope of the invention are as follows:

a) To convey the requests, suggestions and complaints regarding the interface of the learning management system used in the unit to the UQASE.

b) To inform UQASE about the ongoing and completed works related to his / her field of duty.

The diagram showing the necessary support units for the competency-based education quality assurance system model of the inventive system is given below:



The system subject to invention has been developed to be used in associate degree, undergraduate and graduate programs in higher education institutions. In industry, it can be applied within the scope of in-service training. The knowledge and skills required to be taught in the industry in the relevant line of business can be entered into the system, enabling employees to receive training through this system and to be evaluated at the end of the training. Feedback is provided on whether the training provided for the managers and the competencies aimed to be acquired have been acquired by the target audience or not.

CLAIMS

1- The invention is an ahi competency-based education portal that includes the online course module (5), tracking and attendance module (6), and course registration and monitoring module (7) **characterized by comprising;**

- user management and authorization module (1), where authorization is performed in the system for each user through the main server,
- curriculum development module (2), in which the basic parameters of all academic units open in the educational institution (such as the department, department and programs in the Institute / Faculty / Schools) and the curriculum created for each program are automatically transferred to the system via the server,
- content definition module (3) in which the program qualifications and course learning outcomes are defined in the database of the system for academic programs, files and information packages are uploaded for the acquisitions and gains, the subject, objectives and resources of the course are defined,
- measurement and evaluation module (4) that allows the system to analyze the items (difficulty, discrimination, etc.) of the questions as a result of the analysis of the answers entered by the students into the database as a result of the exam conducted on the system and to allow the inappropriate questions to be made passive in order not to be used again by the administrators.

2- The ahi competency-based education portal using user management and authorization module (1), according to Claim 1, **characterized by comprising;** grouping users over the main server and granting them collective authorization.

3- The ahi competency-based education portal using user management and authorization module (1), according to Claim 1, **characterized by comprising;** authorization transition for users with different authorizations on the main server.

4- The ahi competency-based education portal using measurement and evaluation module (4), according to Claim 1, **characterized by comprising;** performing success

analysis of the class or group determined as a result of performing data analysis on the processor during the active period.

5- The ahi competency-based education portal using measurement and evaluation module (4), according to Claim 1, **characterized by comprising;** suggesting exam questions to the department responsible for the exams assigned by the user management and authorization module (1) as a result of the analysis.

6- The ahi competency-based education portal using measurement and evaluation module (4), according to Claim 1, **characterized by comprising;** Providing instructors to create a question pool by entering questions appropriate to their acquisitions into groups categorized for each course on the system.

7- The ahi competency-based education portal using measurement and evaluation module (4), according to Claim 1, **characterized by comprising;** reporting in the database all the evaluations that students have been subjected to throughout their education life, showing the degree of achievement of the qualifications determined at the graduation stage

8- The ahi competency-based education portal using measurement and evaluation module (4), according to Claim 1, **characterized by comprising;** users who are authorized to create a measurement and evaluation module (4) for the courses, and consent to the authorization to conduct exams.

9- The ahi competency-based education portal using user management and authorization module (1), according to Claim 1, **characterized by comprising;** transmitting the entered data to the relevant person or persons over the server in order to control all definitions entered into the system's database and submit it to expert evaluation.

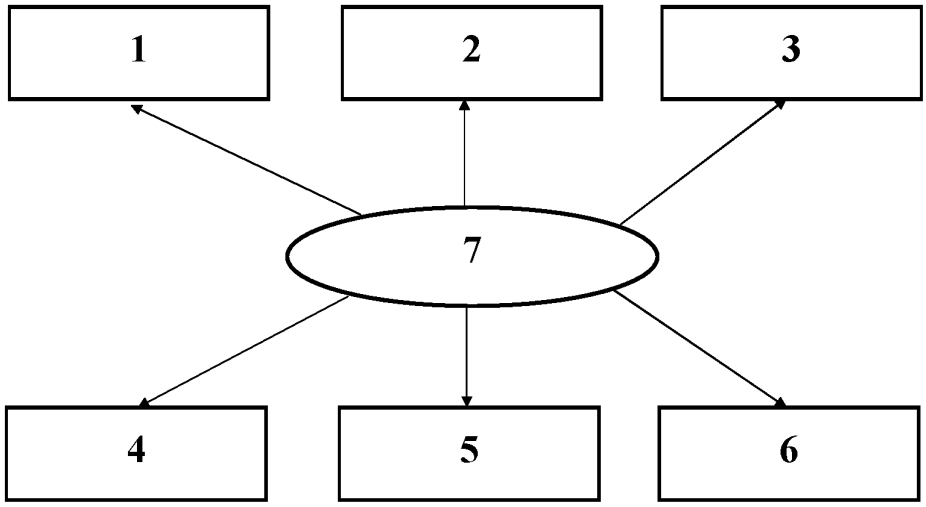


Figure-1

INTERNATIONAL SEARCH REPORT

International application No.
PCT/TR2021/050113

A. CLASSIFICATION OF SUBJECT MATTER

G06F 17/00 (2019.01)i; G06Q 10/06 (2012.01)i; G06F 16/35 (2019.01)i; G06Q 50/20 (2012.01)i; G09B 7/00 (2006.01)i

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G06Q; G06F; G09B

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

PATENW, EPODOC, Turkish National Patent Database, Google Patents (Search Terms: report, analysis, question, suggest, recommend, offer, author, syllabus, curriculum, lesson, course, teacher, instructor, student)

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	CN 110543995 A (UNIV QINGDAO AGRICULTURAL) 06 December 2019 (2019-12-06) the whole document.	1-9
X	US 9626875 B2 (GAL MICHAEL [IL]; KOGAN ARIE [IL]; TIME TO KNOW LTD [IL]) 18 April 2017 (2017-04-18) the whole document.	1-9
A	US 2018350251 A1 (D2L CORP [CA]) 06 December 2018 (2018-12-06) abstract, paragraph [0016].	1-9
A	CN 111882467 A (UNIV HENAN URBAN CONSTRUCTION) 03 November 2020 (2020-11-03) the whole document.	1-9

☐ Further documents are listed in the continuation of Box C. ☑ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"D" document cited by the applicant in the international application
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
24 August 2021	**24 August 2021**

Name and mailing address of the ISA/TR	Authorized officer
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INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
PCT/TR2021/050113

Patent document cited in search report			Publication date (day/month/year)	Patent family member(s)			Publication date (day/month/year)
CN	110543995	A	06 December 2019	NONE			
US	9626875	B2	18 April 2017	CA	2691724	A1	05 February 2009
				AU	2007357074	A1	05 February 2009
				US	2009035733	A1	05 February 2009
				WO	2009016612	A2	05 February 2009
				TW	200917173	A	16 April 2009
				CL	2008002287	A1	19 June 2009
				PE	20090753	A1	15 July 2009
				AR	067789	A1	21 October 2009
				EP	2174313	A2	14 April 2010
				KR	20100042636	A	26 April 2010
				MX	2010001115	A	19 May 2010
				CN	101772792	A	07 July 2010
				JP	2010535351	A	18 November 2010
				RU	2442222	C2	10 February 2012
				IL	202958	A	30 May 2013
				ZA	201000552	B	31 July 2013
				BR	PI0721780	A2	11 February 2014
US	2018350251	A1	06 December 2018	CA	2746039	A1	12 January 2012
				AU	2011203517	A1	02 February 2012
				SG	177849	A1	28 February 2012
				US	2012148999	A1	14 June 2012
				SG	196783	A1	13 February 2014
CN	111882467	A	03 November 2020	NONE			